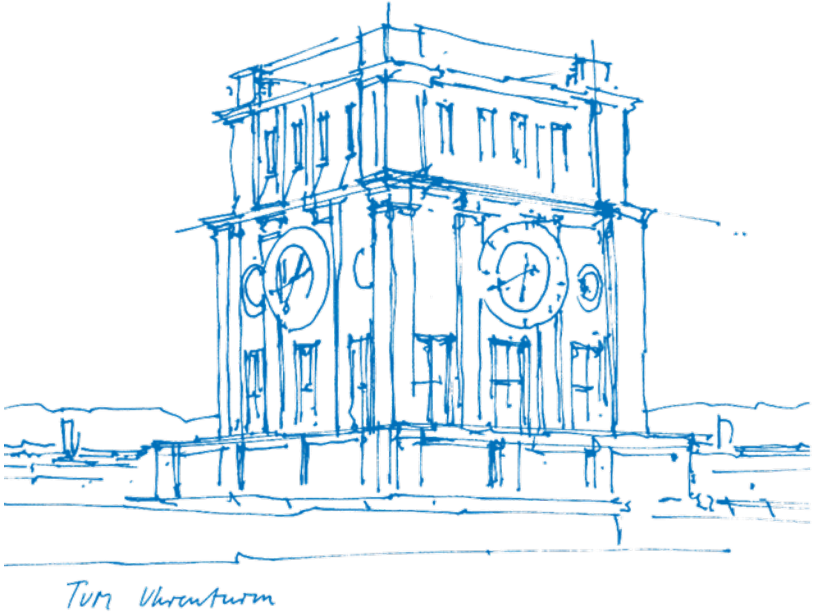
Final Presentation — A DSA-Compliant Tourism RS

David Gallob
Technische Universität München
School of Computation, Information and Technology
Chair of Connected Mobility
Munich, July 6, 2026



Introduction & Motivation

- **The Recommender "Black Box:**
  - Most modern recommender systems (RS) hide their inner workings (e.g. neural network rankings) [1, 2].
  - Causes user distrust, especially in high-stake domains like tourism (travel planning, hotel bookings) [3, 4].
- **Regulatory Requirements: EU Digital Services Act (DSA) [5]:**
  - **Article 27:** Mandates that platforms explain the main parameters of their RS in plain, understandable language.
  - **Article 38:** Mandates that Very Large Online Platforms (VLOPs) offer at least one profiling-free recommendation alternative.
- **Project Goal:** Design and build an interactive, transparent, and DSA-compliant tourism recommender system prototype.

Background: Recommender Systems in Tourism

- **Unique Domain Characteristics [3, 2]:**
  - High financial risk and emotional investment.
  - Infrequent purchasing decisions (trips are planned only once or twice a year).
  - Multi-attribute search spaces (price, location, stars, transport, amenities).
- **Decision Complexity:**
  - Unlike low-cost retail products (e.g. movies, books), users do not buy travel packages on impulse.
  - Users require active comparison tools and want to express precise trade-offs.
  - Trust and perceived control over the algorithm are essential for user acceptance.

Solution Multiple Algorithms including explanations

Two Algorithms with user defined weights/limits

- **5 Attributes for Hotels:**
  - price per night, distance to center, stars, public transport access, user rating
- **Two Algorithms:**
  - **Weighted MAUT:** compensatory approach with user defined weights.
  - **Constraint-Guided:** Focused on hard limit rules and non-compensatory filtering.

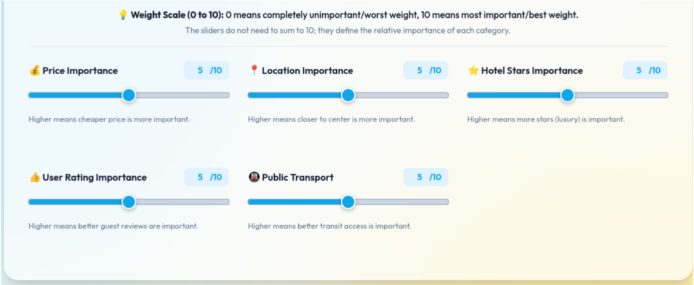


Abbildung: Dashboard

System Architecture & Technical Stack

- **Technology Stack:**
  - React[1] for component-based reactive state rendering.
  - Vite[2] as a fast bundler and build system.
  - Vanilla CSS for a custom, glassmorphic design language.
  - Client-side KaTeX[3] rendering for mathematical formulas.
- **Real-Time Computation Flow:**
  - Eliminates the latency of database queries. Real-time updates occur client-side as users drag the sliders.
  - Caches the hotel calculations, avoiding unnecessary re-renders.
  - Normalization boundaries recalculate dynamically upon city changes.
- **Quality Assurance & Verification:**
  - Standalone recommender engine (`src/recommenderEngine.js`).
  - 60 automated unit tests (`run-tests.js`) verifying edge cases, boundary conditions, and preventing division-by-zero errors.

[1] https://react.dev/

[2] https://vite.dev/

[3] https://katex.org/

Interactive Landing Page for Algorithm Selection

- **Algorithmic State Initialization:**
  - Recommendations, transparency boards, and controls are hidden initially to prevent cognitive overload.
- **Two Strategy Selection Cards:**
  - **Weighted MAUT Card:** Focused on relative importance weighting and compensatory trade-offs.
  - **Constraint-Guided Card:** Focused on hard limit rules and non-compensatory filtering.
- **Strategy Switching:**
  - Simple back button ("← Switch Algorithm") to return to the landing page.
  - Fast toggle tabs in the dashboard header for seamless switching.

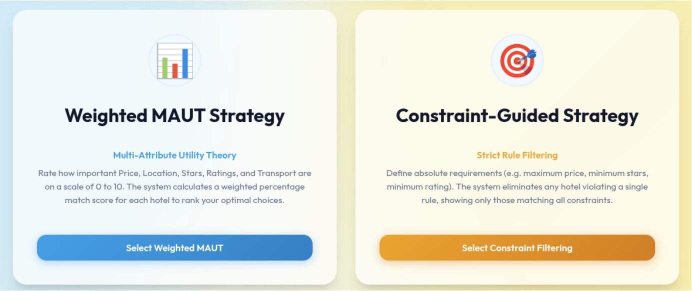


Abbildung: User Weight Sliders (0-10 scale)

Tiered Explainability & Transparency Panels

- **Tier 1: Dynamic Banners (Plain Text)**
  - Dynamic color schemes (Blue for MAUT, Green for Filtering).
  - Brief, plain-language summaries explaining how weights/limits are aggregated.
- **Tier 2: Algorithmic Details (TeX)**
  - Expandable KaTeX mathematical formulas for expert review and full algorithmic disclosure.
- **Tier 3: Step-by-Step Breakdown**
  - On-demand card expansions showing exact computations (Normalization $\times$ Weight).

Abbildung: Step-by-Step Mathematical Breakdown

Demonstration Real-Time Strategy Toggling & Weight Adjustments

Algorithm A: Weighted MAUT (Compensatory)

- **Compensatory Utility Scoring [6]:**
  - Attributes: Price, Location, Stars, Guest Review, Transport.
  - Allows high performance in one attribute (e.g. location) to compensate for low performance in another (e.g. price).
- **Weights on a 0-10 Scale:**
  - Range: 0 (completely unimportant) to 10 (most important).
  - Mathematically normalized by the total sum of user-specified weights:

$$U = \frac{\sum w_i \cdot u_i}{\sum w_i} \times 100\%$$

(w_i: attribute weight [0–10], u_i: normalized utility [0–1], U: match score %)

Weighted Aggregation:

$$U = \frac{\sum_{i=1}^{5} w_i \cdot u_i}{\sum_{i=1}^{5} w_i} \times 100\%$$

Relative Normalization (Price & Distance):

$$u(v) = \frac{v_{\max} - v}{v_{\max} - v_{\min}}$$

Absolute Normalization (Stars & Review/Transport):

$$u_{\text{stars}}(v) = \frac{v-1}{4} \quad \text{and} \quad u_{\text{rating/transport}}(v) = \frac{v-1}{9}$$

Abbildung: User Weight Sliders (0-10 scale)

Normalization Strategies in Weighted MAUT

To aggregate attributes with different units, they are normalized to a $0-1$ scale [7]:

- **Relative Normalization (Cost attributes: Price, Distance):**
  - Normalizes value v relatively based on the active city's min/max values:

$$u(v) = \frac{v_{\max} - v}{v_{\max} - v_{\min}}$$

- Normalizing cost relatively maximizes the contrast between the local hotel options.
- **Absolute Normalization (Benefit attributes: Stars, Rating, Transit):**
  - Normalizes value v on fixed scale bounds to prevent artificial distortion:

$$u_{\text{stars}}(v) = \frac{v-1}{4} \quad \text{and} \quad u_{\text{rating/transport}}(v) = \frac{v-1}{9}$$

Algorithm B: Constraint-Based Filtering

- **Non-Compensatory Logic [3]:**
  - User defines absolute thresholds (e.g. maximum price, minimum stars).
  - A hotel must satisfy all active limits simultaneously to be recommended.
  - Over-performing in one area cannot compensate for a single constraint violation.
- **DSA-Compliant Simplicity:**
  - Unordered subset of matching hotels.
  - Easy-to-understand inclusion/exclusion criteria.
  - Complete profiling-free predictability.

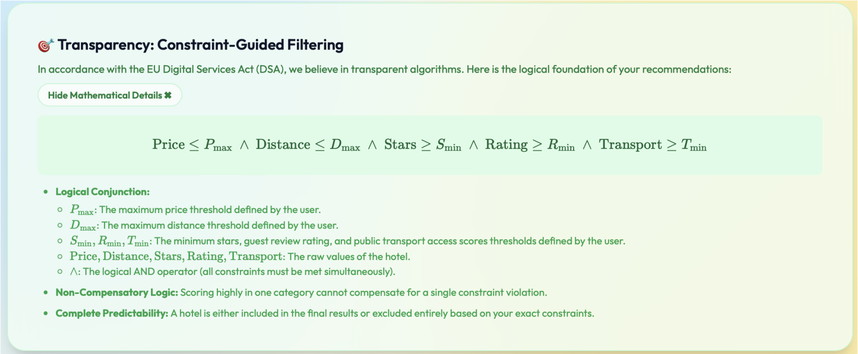


Abbildung: Constraint Filtering Interface

The Satisficing Model of Constraint Filtering

- **Logical Conjunction Formulations:**
  - A hotel is recommended if and only if:

$$\text{Price} \leq P_{\text{max}} \ \wedge\ \text{Distance} \leq D_{\text{max}} \ \wedge\ \text{Stars} \geq S_{\text{min}} \ \wedge\ \text{Rating} \geq R_{\text{min}} \ \wedge\ \text{Transport} \geq T_{\text{min}}$$

- **Psychological Background (Satisficing):**
  - Based on Herbert Simon’s bounded rationality theory (1957).
  - Decision-makers do not seek the mathematically optimal hotel, but rather a "good enough"hotel that meets all minimal conditions.
  - Drastically reduces cognitive load during selection.
  - Perfect for users with rigid conditions (e.g., maximum budget).

UI Redesign & User Experience Challenges

Initial Usability Issues:

- Implicit strategy selection (no clear entry point).
- Cluttered landing state with premature disclosure of complex DSA transparency panels.
- Sliders scaled from $0-100\%$, confusing users into thinking weights must sum to 100%.

Applied Redesign:

- **Interactive Strategy:** landing page.
- Importance scale changed to **0-10 range** to prevent percentage-summing confusion.

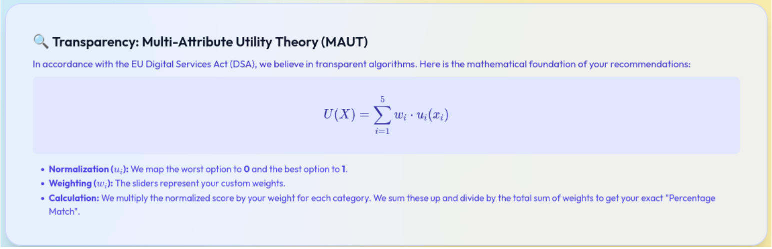


Abbildung: Previous Cluttered Transparency Board

Different Explanation Levels

- **The Explanation Dilemma [8]:**
  - Giving too little explanation makes the system look like a black box.
  - Giving too much explanation causes cognitive overload and user fatigue.
- **Tiered Disclosure (Satisfying Different User Persona):**
  - **Novice Pathway (Banners):** Brief text explanations focusing on the meaning of user controls. Satisfies general transparency.
  - **Expert Pathway (KaTeX Formulas):** Raw equations. Satisfies transparency audits and regulators.
  - **Curious Pathway (Math Breakdown):** Step-by-step breakdown. Explains individual recommendations, boosting user trust.

Future Work: Proposed Hybrid Recommender

Addresses individual limitations of MAUT and Filtering:

- **Limitation of Weighted MAUT:** Compiles complete lists; does not filter out completely irrelevant options (e.g. extremely expensive hotels).
- **Limitation of Constraint Filtering:** Screens items but does not rank remaining candidates (higher cognitive effort).

Proposed Two-Stage Hybrid Approach:

1. **Stage 1: Constraint Filtering (Non-Compensatory screening):** Filter out hotels violating basic requirements (e.g. price $>$ \$250).
2. **Stage 2: Weighted MAUT (Compensatory ranking):** Apply preference weights to rank remaining candidates.

$\Rightarrow$ Maintains complete mathematical transparency, remains profiling-free, and mimics realistic human decision-making.

Proposed User Study & Evaluation Design

- **Hypotheses:**
  - **H1 (Perceived Control):** Users interacting with the Weighted MAUT or Constraint Filtering interfaces report a higher sense of control over recommendations compared to a baseline "black-boxïnterface.
  - **H2 (System Trust):** Exposing the collapsible mathematical breakdown panels increases user trust in the system's recommendations without causing information overload.
  - **H3 (DSA Compliance Impact):** The presence of the global explanation banner and KaTeX formulas increases user comprehension of the algorithm's inner workings.
- **Within-Subjects Laboratory Study ($N = 30$):**
  - **Condition 1 (Baseline):** Traditional black-box hotel list.
  - **Condition 2 (Weighted MAUT):** Slider controls with collapsible breakdowns.
  - **Condition 3 (Constraint Filtering):** Threshold sliders.

Summary & Conclusion

- **Successfully Completed Deliverables:**
  - Developed an interactive, responsive tourism recommender SPA (React + Vite).
  - Redesigned selection flow with a strategy landing page to prevent early transparency board clutter.
  - Configured sliders on an intuitive 0-10 scale to eliminate weight-summing misunderstandings.
  - Integrated tiered explainability panels (plain-language summaries, LaTeX formulas, math breakdowns).
- **Key Takeaway:**
  - Demonstrates that EU DSA compliance (Article 27 & 38) can be harmonized with rich user controls, mathematical transparency, and premium UI/UX aesthetics.

Thank you for your attention!

Questions?

References I

[1] Gediminas Adomavicius und Alexander Tuzhilin. „Toward the next generation of recommender systems: A survey of the state-of-the-art and possible extensions“. In: *IEEE Transactions on Knowledge and Data Engineering* 17.6 (2005), S. 734–749. DOI: 10.1109/TKDE.2005.99.

[2] Joan Borras, Antonio Moreno und Aida Valls. „Intelligent tourism recommender systems: A survey“. In: *Expert Systems with Applications* 41.16 (2014), S. 7370–7389. DOI: 10.1016/j.eswa.2014.06.007.

[3] Francesco Ricci. „Travel recommender systems“. In: *Recommender Systems Handbook*. Springer, 2011, S. 527–556. DOI: 10.1007/978-0-387-85820-3_17.

[4] Bart P. Knijnenburg u. a. „Explaining the user experience of recommender systems“. In: *User Modeling and User-Adapted Interaction* 22.4-5 (2012), S. 441–504. DOI: 10.1007/s11257-011-9118-4.

[5] European Parliament and Council of the European Union. *Regulation (EU) 2022/2065 of the European Parliament and of the Council of 19 October 2022 on a Single Market For Digital Services and amending Directive 2000/31/EC (Digital Services Act)*. http://data.europa.eu/eli/reg/2022/2065/oj. 2022.

[6] Christian Schmitt, Dietmar Dengler, Michael Bauer u. a. „The MAUT-Machine: An Adaptive Recommender System“. In: *Proceedings of the ABIS Workshop*. Hannover, Germany, 2002.

References II

[7] Ralph Keeney, Howard Raiffa und David Rajala. „Decisions with Multiple Objectives: Preferences and Value Trade-Offs". In: *Systems, Man and Cybernetics, IEEE Transactions on* 9 (Aug. 1979), S. 403–403. DOI: 10.1109/TSMC.1979.4310245.

[8] Nava Tintarev und Judith Masthoff. „A Survey of Explanations in Recommender Systems". In: Mai 2007, S. 801–810. ISBN: 978-1-4244-0832-0. DOI: 10.1109/ICDEW.2007.4401070.

[9] Mohamed Elyes Ben Haj Kbaier, Hela Masri und Saoussen Krichen. „A Personalized Hybrid Tourism Recommender System". In: *2017 IEEE/ACS 14th International Conference on Computer Systems and Applications (AICCSA)*. 2017, S. 244–250. DOI: 10.1109/AICCSA.2017.12.

[10] Joy Lal Sarkar u. a. „Tourism recommendation system: a survey and future research directions". In: *Multimedia Tools and Applications* 82.6 (2023), S. 8983–9027. DOI: 10.1007/s11042-022-12167-w.

[11] Sean M. McNee, John Riedl und Joseph A. Konstan. „Being accurate is not enough: How accuracy metrics have hurt recommender systems". In: (2006), S. 1097–1101. DOI: 10.1145/1125451.1125659.

[12] Swearingen Sinha, Kirsten Medhurst und Rashmi Sinha. „Beyond Algorithms: An HCI Perspective on Recommender Systems". In: (Sep. 2001).

Backup Slides

Traditional Recommender System (RS) Strategies

- **Collaborative Filtering (CF):**
  - Predicts preferences based on ratings from similar users.
  - *Limitations in Tourism:* Cold start problems (new hotels/infrequent travelers) and lacks explanatory power.
- **Content-Based Filtering (CB):**
  - Recommends items matching a user profile built from historical behavior.
  - *Limitations in Tourism:* Poor adaptability to sudden context changes (e.g. business trip vs. family vacation).
- **Hybrid Systems [9, 10]:**
  - Combine CF and CB. However, they remain mathematical black boxes to the end-user.
  - Hard to explain, violating transparency guidelines.

HCI: The Importance of User Control & Trust

- **Being Accurate is Not Enough [11]:**
  - Traditional metrics focus solely on algorithmic accuracy (e.g., RMSE).
  - Ignores user factors like transparency, control, and serendipity.
- **Beyond Algorithms [12]:**
  - Interactive systems should give users a direct steering wheel.
  - Sliders and constraints shift the role of the user from a passive consumer to an active co-creator.
- **Explaining the User Experience [4]:**
  - Combining user control with visual explanations leads to significantly higher trust, perceived recommendation quality, and system satisfaction.

Proposed User Study Evaluation Metrics

To scientifically test the hypotheses, standardized metrics will be gathered:

- **System Usability Scale (SUS):**
  - 10-item Likert scale questionnaire to evaluate overall system usability.
- **ResQue (Recommender Systems Quality of Experience) [4]:**
  - Focuses on user-centric quality: perceived quality, system trust, transparency, and intent to use.
- **NASA Task Load Index (NASA-TLX):**
  - Focuses on subjective cognitive workload.
  - Measures: Mental demand, physical demand, temporal demand, performance, effort, and frustration level. Helps verify if exposing math equations causes overload.